

Total No. of Questions : 8]

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Roll No

CS-7002-CBGS

B.E. VII Semester

Examination, December 2020

Choice Based Grading System (CBGS)

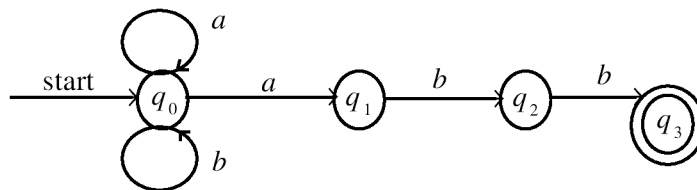
Compiler Design

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) What is a front end and back end of a compiler? What are the advantages of breaking up the compiler functionality into these two distinct stages?
b) Why the analysis portion of a compiler is normally separated into lexical analysis and parsing phasor? Explain tokens, Patterns and lexemes with an example.
2. a) Obtain a DFA equivalent to the NFA given:



- b) Differentiate between NFA and DFA under following points:
Power, Transition function, Time complexity, Supremacy, Transition Diagram and Transition Table.

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3. Consider the following grammar:-
 $E \rightarrow E+T/T$
 $T \rightarrow TF/F$
 $F \rightarrow F*/a/b$
i) Construct the SLR parsing table for this grammar.
ii) Construct the LALR parsing table.
4. a) Why is it necessary to eliminate left recursion in top down parsing? Consider predictive parsing table for:
 $S \rightarrow (L) / a$
 $L \rightarrow L, s/s$
b) Differentiate between simple LR parser and canonical LR parser.
5. Explain implicit type conversion and explicit type conversion with example.
6. a) What is an intermediate code? What are the benefits of intermediate code generation ? How can three address code be implemented in a compiler? Describe quadruple.
b) Define backpatching and semantic rules for Boolean expression. Derive the three address code for the following expression $P < Q$ and $R < S$ and $T < U$.
7. a) Explain in brief the various issues of design of a code generator.
b) Explain the basic block and control flow graph.
8. a) Write short notes on:
i) Global Data Flow Analysis
ii) Loop Optimization
b) Explain the concept of dead code elimination.
